2.3 Step 1: Comprehensive multi-temporal landslide inventory

2.3.1 Objective

The goal of this step is to identify where and when landslides occurred, the types of slope failure processes, the level of ground deformation activity, the history and origin of the landslides (causes and triggers). Concerning the latter point, understanding the role of human disturbances in the slope failure is important.

2.3.2 Method

The landslide inventory was built from a careful and detailed 3D (elevation exaggeration of 1) visual interpretation of Google Earth images. All images used in the analysis are of very-high spatial resolution, ranging from 30 to 60 cm. The images in Google Earth are provided by either © DigitalGlobe or © CNES/© Airbus and they were captured between 2000 and 2021. The analysis of Google Earth images has proven to be a successful and reliable method to map landslides (Fisher, et al., 2012). The reliability of the approach has been demonstrated by Depicker et al. (2020a; 2020b) for the Lake Kivu region, including the western part of Rwanda. The satellite images were analyzed in parallel with the photographs taken in the field.

The term landslide encompasses a large range of gravity-driven mass movements, that differ by their failure mechanisms, size, depth, velocity and the material mobilised (Lu and Godt, 2013; Hungr et al., 2014). We used the updated Varnes' classification (Hungr et al. 2014) for defining the landslide typology; the classification schemes being the most widely used. Defining the landslide typology is a key step when predicting the associated risk as the variety of the slope processes (size, speed, origin, etc.) is large and hence the types of impacts (Corominas et al., 2014).

The landslide depth is an important element when analysing landside causes and triggers; the occurrence of shallow landslides being for instance much more sensitive to disturbances of the landscape surface and rainfall conditions than deep-seated landslides (Sidle and Bogaard 2016). For example, the removal of trees, due to either human or natural causes, decreases the slope stability through the alteration of hydrological and geotechnical conditions, such as the loss in soil cohesion due to tree root decay (Sidle et al., 2006; Sidle and Bogaard, 2016). Deep-seated landslides, both rapid and slow-moving, are also controlled, like the shallow landslides, by water routing through the regolith (see Text Box 1). Slow, deep-seated landslides (e.g., slides and earthflows; Figure 20) typically require an extended period of water recharge to initiate movement, while rapid, deep-seated failures may initiate by either direct response to individual storms or prolonged water inputs (Sidle and Bogaard, 2016). The depth of the surface of rupture of shallow landslides is usually defined in the range of 2-5 m (Keefer, 1984; Mackey and Roering, 2011; Sidle and Bogaard, 2016). Here, we considered landslides to be shallow when their estimated depth is < 5 m. We estimated the relative depth of the landslides visually analysing the shape and size of the landslide scarp and deposits in Google Earth imagery. We have validated such an visual analysis through in situ field observations in similar context in Rwanda and the Kivu Rift (Depicker et al., 2020a, 2020b). The landslides occurring in mining and quarrying sites were all classified as quarrying landslides, regardless of their depth.